

DAR ES SALAAM INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER STUDIES

NTA LEVEL 8



BLOCKCHAIN-BASED NGO DONATION MANAGEMENT SYSTEM

MODULE NAME: PROJECT REALIZATION

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COURSE: BENG22COE

CANDIDATE NAME: NASIBU Y. ISAKA

REGISTRATION NUMBER: 2106307225519

SUPERVISOR'S NAME: DR GUSTAPH SANGA

[PENDING: SUBMISSION DATE]

BLOCKCHAIN-BASED NGO DONATION MANAGEMENT SYSTEM

Changia

NASIBU Y. ISAKA

2106307225519

A project report submitted in partial fulfilment of the requirements for
the award of NTA Level 8 in the Department of Computer Studies of
the Dar es Salaam Institute of Technology.

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[PENDING: SUBMISSION DATE]

Declaration

I, **Nasibu Y. Isaka**, registration number **2106307225519**, declare that this report is my own original work and that it has not been presented, and will not be presented, to any other institution for a similar or any other award.

Signature: _____

Date: _____

Certification

The undersigned certifies that he has read and hereby recommends for acceptance by the Dar es Salaam Institute of Technology a project report titled **Blockchain-Based NGO Donation Management System**, submitted in partial fulfilment of the requirements for the award of NTA Level 8 in the Department of Computer Studies.

Dr Gustaph Sanga

Supervisor

Signature: _____

Date: _____

Dedication

Acknowledgements

Abstract

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List of Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
CSV	Comma Separated Values
DFD	Data Flow Diagram
DIT	Dar es Salaam Institute of Technology
ERD	Entity Relationship Diagram
FR	Functional Requirement
JWT	JSON Web Token
NFR	Non-functional Requirement
NGO	Non-Governmental Organization
NTA	National Technical Award
ORM	Object Relational Mapping
RBAC	Role Based Access Control
RPC	Remote Procedure Call
SDLC	Software Development Life Cycle
SRS	Software Requirement Specification
TZS	Tanzanian Shilling
UAT	User Acceptance Testing
UML	Unified Modeling Language

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C	Source Code (selected excerpts)
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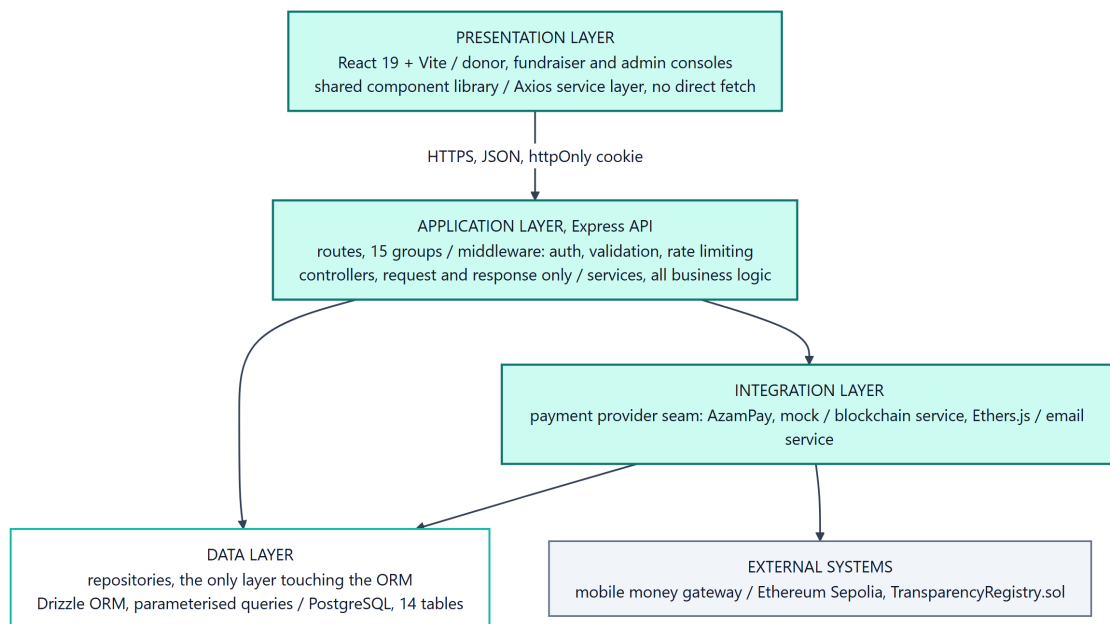


Figure 6.1. System architecture, layered view

6.3 Data Modeling

6.3.1 Context Diagram

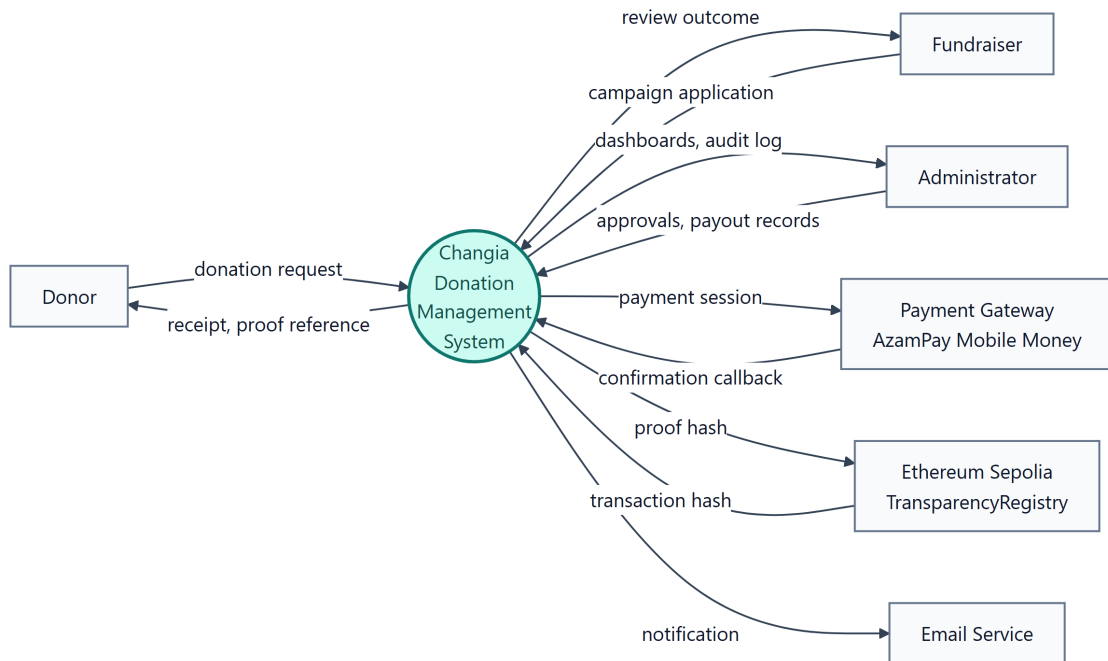


Figure 6.2. Context diagram, DFD level 0

Data Flow Diagrams

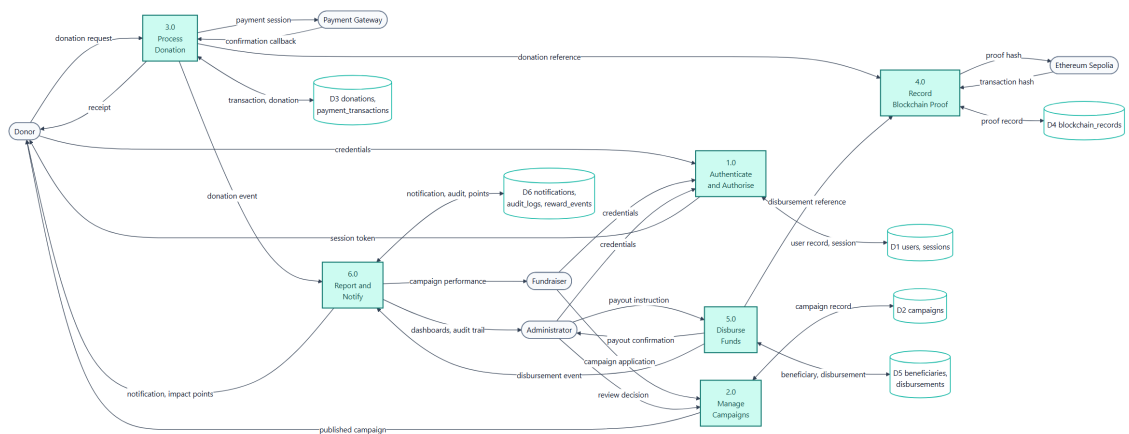


Figure 6.3. Data flow diagram, level 1

6.4.1 Database Design

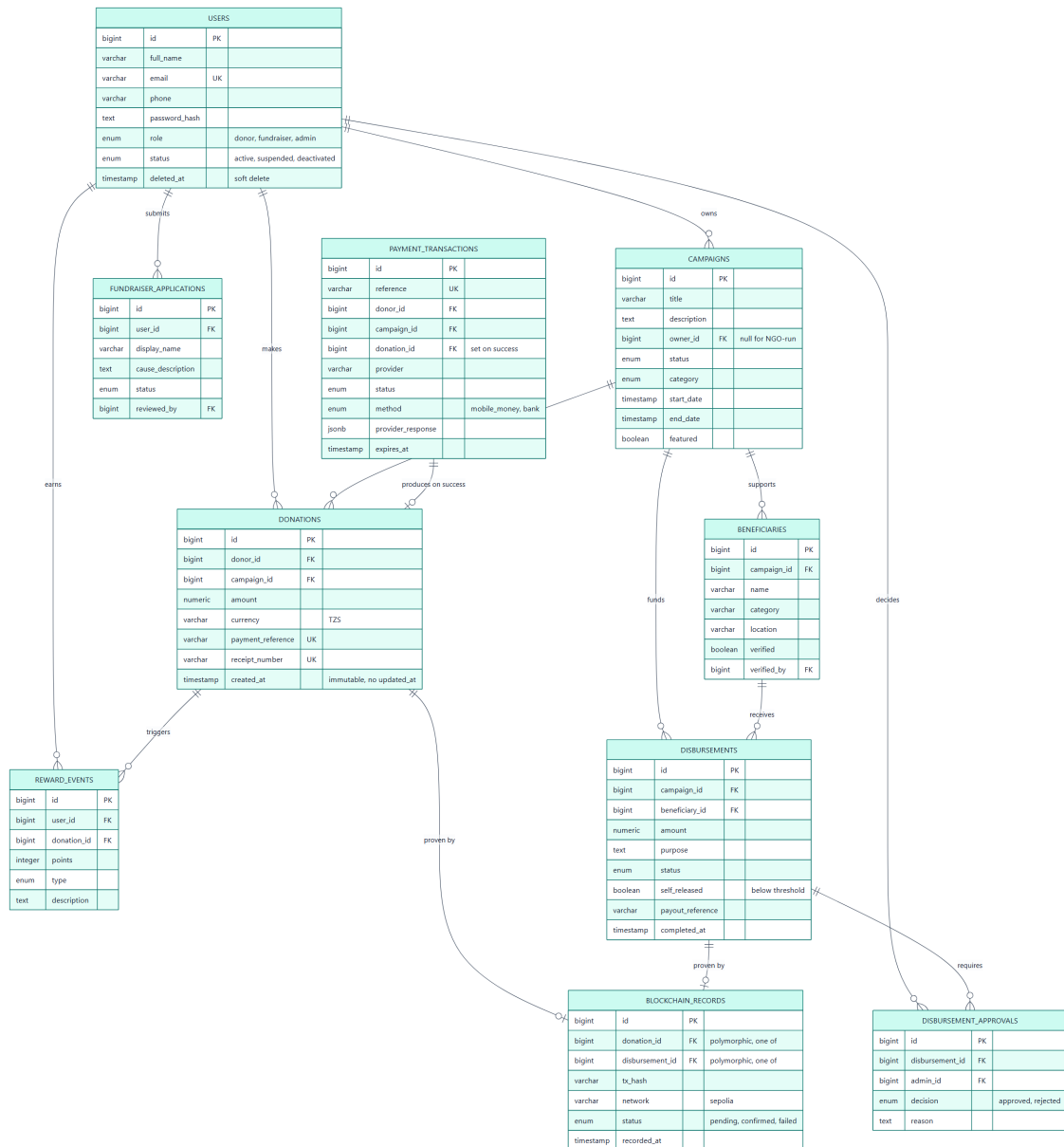


Figure 6.4. Entity relationship diagram

6.5 Process Modeling

6.5.1 UML Models

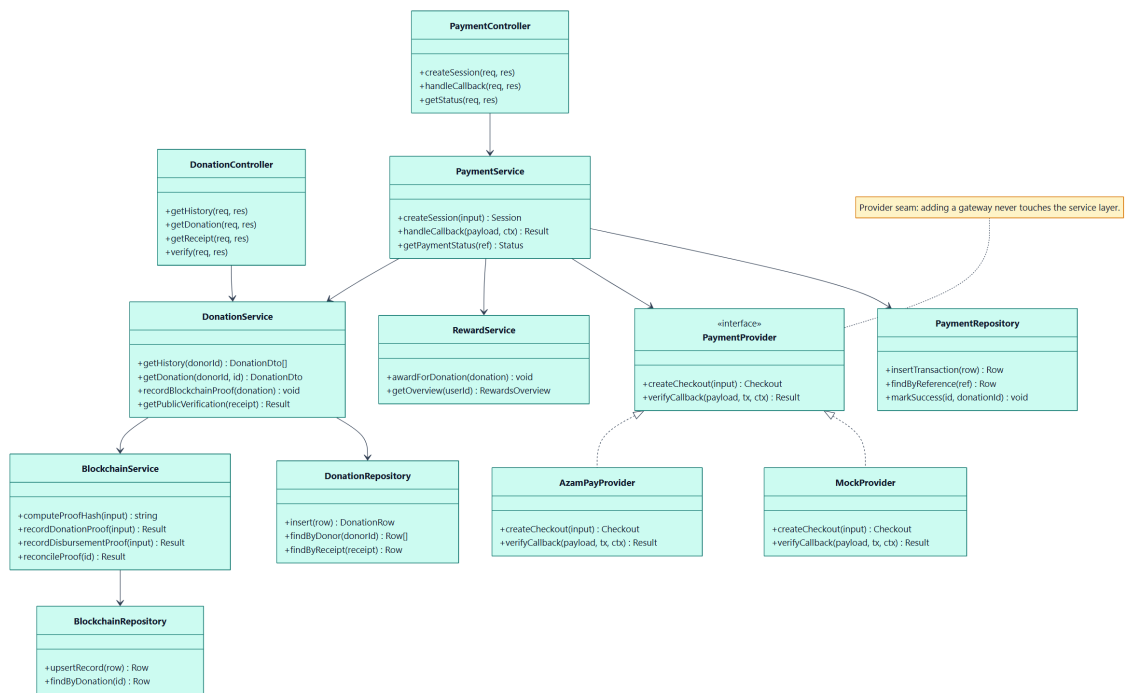


Figure 6.5. Class diagram, backend service layering

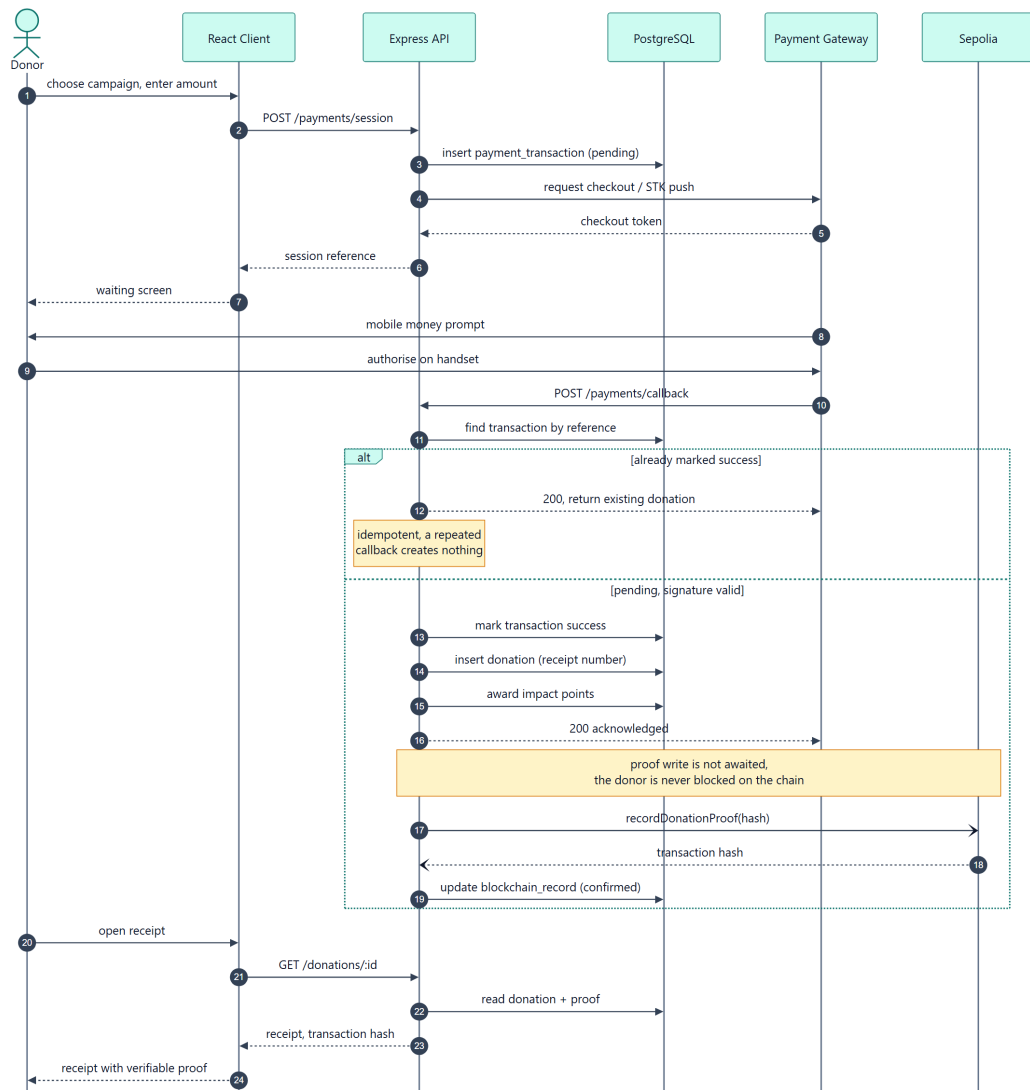


Figure 6.6. Sequence diagram, donation and proof recording

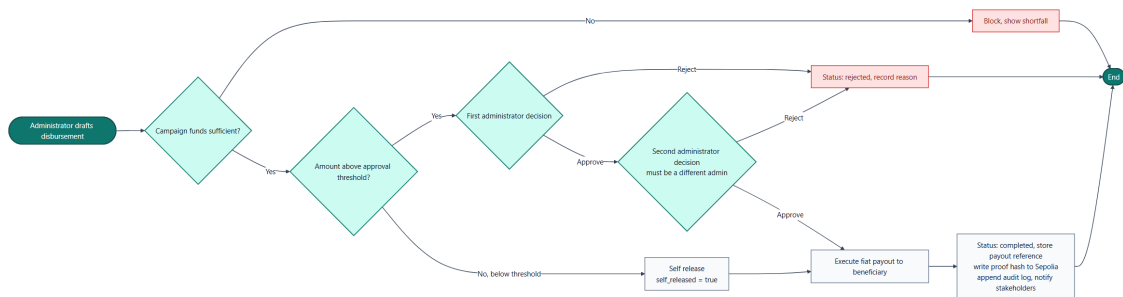


Figure 6.7. Activity diagram, disbursement dual approval

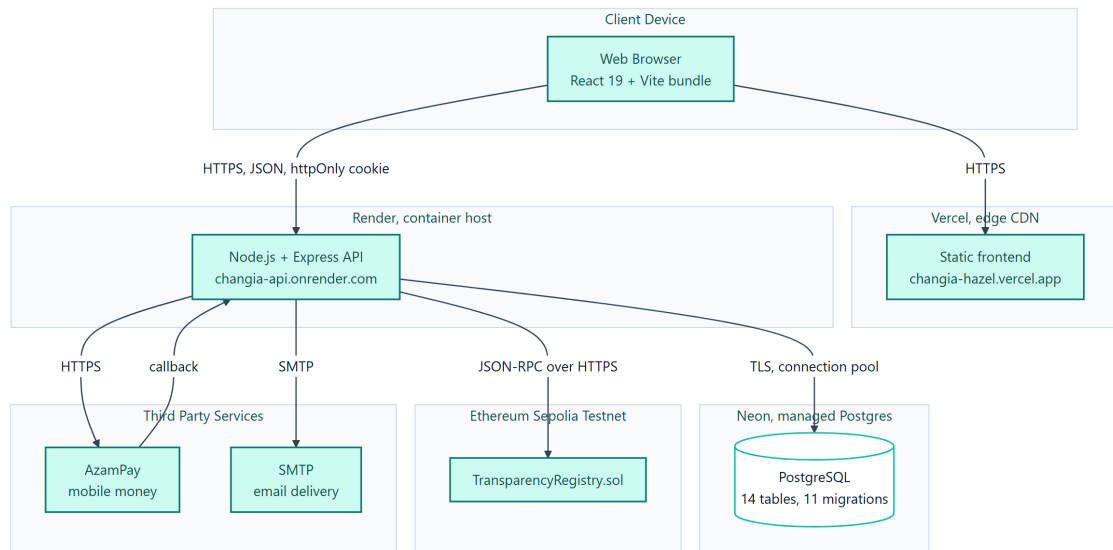


Figure 6.8. Deployment diagram

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